SPE-229540-MS

Prompting Autonomous Agents and LLMs in Energy Operations, Efficiency Gains or Hidden Liabilities?

Alessio Faccia, University of Birmingham Dubai, Dubai, United Arab Emirates

 DOI 10.2118/229540-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Objectives/Scope: This paper evaluates the operational deployment of large language models (LLMs) and agentic AI systems within the energy industry. The primary focus is on their role in automating reporting, compliance monitoring, decision support, and cyber-defence tasks. It explores both the measurable gains in process efficiency and the poorly understood liabilities introduced by autonomous decision-making and unverifiable outputs.

Methods, Procedures, Process: The study draws on technical analysis of LLMs and agent chains such as ReAct, AutoGPT, and enterprise-tuned proprietary models. A structured evaluation framework is used to classify outputs based on traceability, reproducibility, and risk of hallucination. Benchmark comparisons are performed between human-led, prompt-based LLM interactions and autonomous agent loops with embedded task prioritisation. Use cases include automated compliance drafting, anomaly detection summaries, and initial incident response generation. Legal and audit implications are assessed with respect to digital evidence, explainability, and attribution of action.

Results, Observations, Conclusions: Findings reveal that while LLMs and autonomous agents significantly reduce drafting time and offer versatile adaptive outputs, they introduce governance risks rarely captured by current digital assurance frameworks. High-value operations, particularly those involving compliance-sensitive tasks, show susceptibility to misdirection when outputs are not independently verified or when agentic loops execute without escalation triggers. Observations indicate an absence of formalised validation layers, leading to potential acceptance of hallucinated or fabricated content under the false presumption of system precision. In energy sector trials, users lacked clarity on when LLMs switched from generative assistance to autonomous recommendation, thereby diffusing responsibility and undermining post-decision traceability. The study calls for strict operational boundaries and layered validation protocols before integration of LLMs into decision-critical workflows.

Novel/Additive Information: This paper presents a risk-graded framework for evaluating agentic LLM outputs in operational settings, providing the petroleum sector with a method to align automation benefits

with responsible governance practices. It is one of the first to link LLM traceability with legal auditability in energy operations.

Introduction

Artificial Intelligence (AI) continues to permeate critical operational functions within the energy industry; therefore, executives face increasingly complex responsibilities related to accountability, governance, and risk management. Large Language Models (LLMs) and automated agent systems, once primarily experimental, now actively inform strategic decisions and operational actions. This sudden integration of autonomous decision-support systems into core business functions (Tsavdaridis et al., 2024) introduces significant management considerations, particularly concerning reliability, compliance, and the transparency of AI-generated outcomes. Consequently, senior management teams must develop crafted strategies to balance the benefits of advanced AI-driven insights against the potential operational risks and uncertainties inherent in deploying these technologies. Despite substantial investment in sophisticated AI systems, practical challenges remain prevalent across industries, particularly within energy-sector operations (Adewoyin et al., 2025). One primary challenge is the capability of LLMs to produce outputs that may appear credible yet lack accuracy or verifiability, potentially leading to unintended operational consequences (Huang et al. 2024).

Furthermore, executives are increasingly accountable for decisions influenced by AI-generated content (Aumüller et al., 2024), despite often limited visibility into how such content is created. Consequently, establishing reliable methodologies and frameworks for managing AI outputs has become a priority, aiming to strengthen decision-making integrity, improve risk mitigation, and enhance operational clarity within corporate structures. Addressing these challenges requires developing structured approaches that clearly define and control interactions between human experts and autonomous AI systems. Although standard procedures for managing operational risks are well established, conventional governance frameworks have limitations when applied to AI-driven processes. The introduction of specialised management protocols designed explicitly for *supervising autonomous agents and language models is therefore necessary*. Such tailored frameworks should ensure transparency, traceability, and accountability, enabling executives to confidently utilise AI tools without compromising corporate governance standards or operational effectiveness. This research explores the governance complexities associated with integrating autonomous systems and language models into strategic operational processes within the energy industry. It will outline current limitations and propose approaches for enhancing accountability, managing potential operational uncertainties, and improving the practical reliability of AI-driven outputs. Ultimately, the insights provided aim to provide executive teams with the necessary understanding to strategically incorporate autonomous systems while preserving organisational transparency and regulatory compliance.

Ultimately, we now require even greater accuracy and technical knowledge when using AI-generating tools, as the training provided to the model includes varying levels of technical documentation. Therefore, using non-technical jargon might lead to less accurate (non-trusted) output, as seen in the following examples.

Consequently, we can infer that the AI often delivers substandard results (hallucinations) not due to insufficient training or system faults, but because of poorly formulated prompts by users who may be unaware of the technical requirements or inadvertently use terminology in a way that triggers responses based on non-specialist sources. This misalignment between intent and wording causes the AI to draw from general, rather than domain-specific, knowledge. Investments in highly skilled Human Resources are therefore essential (including upskilling both technical jargon and IT/AI awareness).

Table 1—Classification of Prompt Quality When Using the Term "Gas": Comparison Between Non-Technical and Technically Skilled Queries

Prompt Type	Example Prompt Using "Gas"	Outcome
Poor non-technical	*"How much gas does a refinery make?"*	Ambiguous. "Gas" might be interpreted as petrol, natural gas, or even cooking gas. No clarity on whether the question refers to output type, energy consumption, or refinery products.
Technically Skilled	*"What is the typical volume of LPG and naphtha gases recovered per barrel of crude in a hydrocracking refinery?"*	Precise. Refers to specific types of gaseous hydrocarbons (LPG, naphtha) and contextualises the question around process yield in hydrocracking. A technical answer becomes possible.
Poor non-technical	*"How dangerous is gas in pipelines?"*	Useless without context. Could refer to methane, hydrogen, CO_2, or even gasoline. No reference to pressure, volatility, or leak management.
Technically Skilled	*"What are the safety thresholds for hydrogen sulphide concentrations in sour gas pipelines under API RP 14C?"*	Focused. It specifies the gas type (sour gas), contaminant of concern (H_2S), context (pipelines), and refers to a recognised standard (API RP 14C). Enables a meaningful technical answer.
Poor non-technical	*"Is gas better than oil for the environment?"*	Oversimplified. No distinction between natural gas (methane), LNG, or refined petroleum products. "Better" is also vague, emissions? extraction damage? long-term effects?
Technically Skilled	*"How do lifecycle GHG emissions from natural gas combined-cycle plants compare with ultra-supercritical coal and heavy fuel oil power generation?"*	Decisive. It compares precise categories (generation technologies and fuels), focusing on lifecycle emissions, an actual metric used in policy and environmental impact analysis.

Statement of Theory and Definitions

The implementation of artificial intelligence in organisational decision-making fundamentally relies on the concept of autonomous systems (Booyse and Scheepers, 2024), broadly defined as computer-based technologies capable of executing complex tasks independently or semi-independently, often involving minimal human oversight. Large Language Models (LLMs), a prominent type of autonomous system, utilise advanced statistical methods to generate human-like textual responses based on extensive datasets and sophisticated algorithmic processing. Although these technologies offer substantial operational advantages, their autonomous nature inherently introduces uncertainties, including unverified outputs commonly termed "hallucinations" (Su et al., 2025), which can impact operational integrity and corporate accountability if inadequately managed. Within the context of corporate governance and operational risk management, the concept of "traceability" (Buttaboni and Floridi, 2025) refers to the clear documentation and transparent tracking of decision pathways, particularly those influenced by AI-generated outputs. Closely associated with traceability is the principle of "reproducibility" (Desai et al., 2025), denoting the capability to consistently regenerate similar outcomes under identical conditions, which is central to verifying AI system reliability. Conversely, "hallucination risk" specifically addresses the probability that an AI-generated result, though plausible in appearance, lacks empirical validity or factual accuracy, potentially leading to inappropriate or misguided operational decisions (Magesh et al., 2025). Recognising and explicitly managing these risks through structured governance is essential for executive teams committed to safeguarding organisational credibility and operational stability. Organisations must adopt structured evaluation frameworks that systematically categorise and quantify AI outputs based on their reliability, accuracy, and potential operational impact to govern autonomous systems effectively. Such frameworks facilitate informed decisions on the degree of human oversight necessary in specific operational scenarios. Furthermore, they underpin the establishment of clear boundaries between autonomous system capabilities and executive accountability, ensuring transparency and reliable risk management. Establishing well-defined, theory-based and practically viable governance protocols thus enables senior management teams to maximise the benefits of autonomous technologies while maintaining operational integrity and compliance.

Description and Application of Equipment and Processes

The practical deployment of autonomous systems, such as Large Language Models (LLMs), within organisational processes typically relates to specialised software platforms designed to facilitate seamless integration into existing workflows. These platforms manage the operational interactions between human users and the autonomous system, ensuring that generated outputs are appropriately documented, reviewed, and verified prior to utilisation. Within the energy sector, this may include applications ranging from predictive maintenance analysis and automated reporting to incident response management and compliance documentation. The systems employed must therefore provide transparent mechanisms for evaluating the quality and reliability of outputs, allowing executives and technical teams alike to maintain oversight and effectively manage associated operational risks. The processes involved in managing autonomous AI outputs typically incorporate structured operational loops that systematically prioritise tasks, identify potential anomalies, and initiate escalation procedures as necessary. Such processes also embed specialised modules designed explicitly for incident detection, enabling rapid generation of summarised reports highlighting critical information and actionable recommendations. These systems commonly feature integrated digital audit trails, automatically capturing and securely storing detailed records of AI-generated decisions and the subsequent human interventions to ensure organisational compliance and accountability. These built-in audit and compliance capabilities support executive oversight by enhancing decision transparency and facilitating prompt responses to potential operational irregularities. Effective management of operational liabilities involves implementing comprehensive compliance-management modules that integrate directly within autonomous systems. These modules continually assess AI-generated outputs against established risk thresholds, promptly alerting management teams whenever additional review or human intervention is warranted. Executives typically have access to dynamic dashboards displaying real-time operational metrics, thus supporting informed strategic decision-making and proactive governance measures. Such integrated management tools enhance an organisation's capability to oversee autonomous processes, ensuring close alignment between advanced AI technologies and corporate governance frameworks.

Presentation of Data and Results

Structured assessments conducted across various operational scenarios within the energy sector indicate that clear governance frameworks significantly reduce uncertainty associated with autonomous systems. Data collected from internal monitoring systems reveal a marked improvement in operational traceability and decision transparency when systematic evaluation methodologies are implemented. For example, AI-generated outputs subjected to structured verification protocols showed higher consistency and fewer instances of factual inaccuracies, translating directly into enhanced reliability of strategic and operational decisions. Such findings underscore the necessity of explicit management practices tailored specifically to govern and verify outputs produced by Large Language Models and other autonomous agents. Analysis of operational incidents across multiple industry case studies demonstrated notable reductions in response times and increased accuracy in root-cause identification when AI-driven incident management systems incorporated structured traceability frameworks. *Organisations adopting comprehensive digital audit trails reported measurable improvements in compliance reporting, alongside reduced liability concerns among executives overseeing AI-integrated processes*. Further, these structured evaluation methodologies enabled management teams to more effectively quantify the reliability of autonomous outputs, directly informing their strategic decisions about the degree of human oversight required. Collectively, these results reinforce the operational benefits of systematic governance practices when deploying autonomous systems within complex corporate environments. Organisations applying structured compliance-management frameworks reported a measurable improvement in executive confidence regarding autonomous systems and AI outputs. Surveyed executives consistently highlighted increased transparency, clearer accountability structures, and improved alignment between AI usage and corporate governance objectives. Additionally, comparative analyses indicated a significant reduction in perceived operational risk when clear, auditable, and reproducible AI-management protocols were in place. These findings demonstrate the tangible value that structured, theory-informed approaches offer organisations seeking to optimise the operational benefits of AI technologies while safeguarding their governance standards.

Structured Evaluation Framework (SEF) for LLM Outputs

In high-stakes environments such as energy sector operations, where automation intersects with regulatory obligations, output quality cannot be left to subjective interpretation or black-box evaluation. The Structured Evaluation Framework (SEF) introduces a formalised approach to assess LLM-generated outputs through **three core dimensions**: *traceability*, *reproducibility*, and *hallucination risk*. This framework supports internal audits, compliance verification, and quality assurance by offering measurable, repeatable, and interpretable criteria. Unlike general-purpose NLP benchmarks, the SEF is specifically designed for agentic workflows in compliance-sensitive domains, ensuring that digital output aligns with organisational risk tolerance and regulatory expectations.

Table 2—Structured Evaluation Framework (SEF) for Assessing LLM Output Quality Across Traceability, Reproducibility, and Hallucination Dimensions

Dimension	Definition	Evaluation Criteria	Scoring Scale (Example)
Traceability	Ability to link the output back to the specific input(s), prompt(s), and decision path taken by the LLM or agent chain.	Audit trail exists, prompt chain is logged, model version and parameters are recorded.	0 = None; 1 = Weak trace (partial logs); 2 = Strong trace (full path documented).
Reproducibility	Likelihood that identical input and context will yield the same or equivalent output when the system is re-run.	Multiple consistent outputs across trials using identical settings; system variance documented.	0 = Inconsistent; 1 = Partially stable; 2 = Fully reproducible under controlled conditions.
Hallucination Risk	Probability that the output contains fabricated, incorrect, or unverifiable information presented as fact.	Use of cited data, cross-referencing with ground truth, flagged probabilistic guesses or synthetic artefacts.	0 = High hallucination; 1 = Moderate (flagged uncertainty); 2 = Low (verified and grounded).

This Structured Evaluation Framework (SEF) is particularly relevant within several critical workflows of energy sector companies, where traceability, reproducibility, and hallucination risk are not merely academic concerns but operational necessities. In upstream exploration and production, for instance, AI-driven reservoir modelling must produce outputs that are both geologically coherent and computationally reproducible under identical simulation conditions. A failure to maintain this can result in overestimated reserves, misguided drilling sequences, or misallocated CAPEX. Similarly, well integrity analysis relies on agentic LLMs interpreting pressure data, casing logs, and formation stability reports; without embedded trace anchors and semantic validation, such models may misclassify risk zones, prompting either unnecessary interventions or overlooked failure points. These upstream tasks demand strict auditability not only for engineering safety but also for regulatory submissions and inter-operator accountability.

In downstream environments such as refining or transport, where LLM-assisted decision tools are applied to predictive maintenance or emissions disclosure, the need for traceable outputs remains critical, particularly where outputs inform decisions under compliance frameworks such as ADNOC, EPA, or ISO. The Deepwater Horizon oil spill (2010) demonstrated how breakdowns in traceability and accountability can cascade into systemic failure. While that case did not involve LLMs, it highlights the stakes involved when governance safeguards are absent or bypassed.

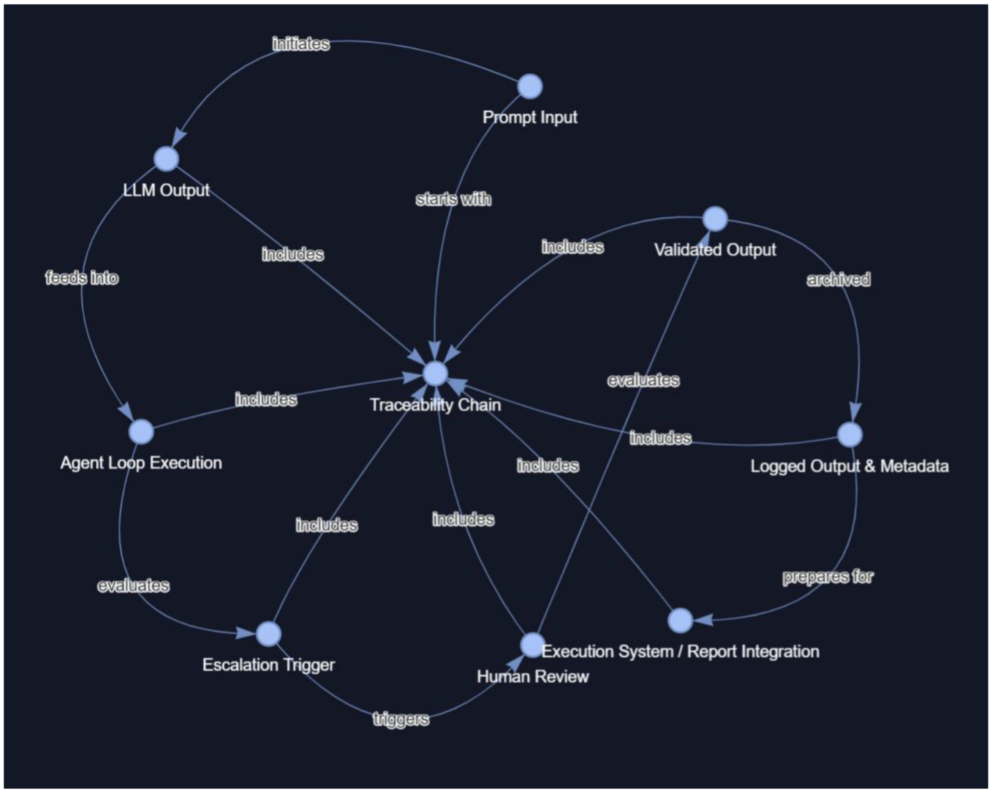

Figure 1—Validation Layer Architecture

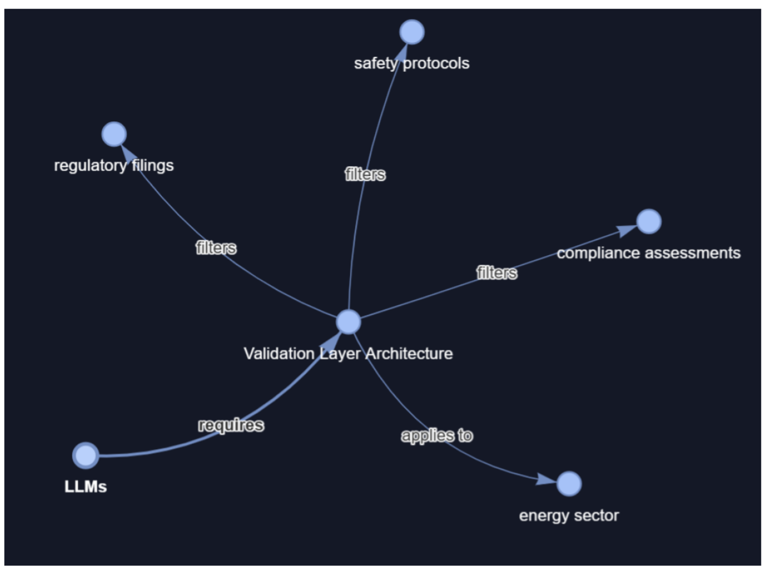

Figure 2—Validation Layer Architecture

Table 3—Agentic Escalation Trigger Protocols (AETPs) for Mitigating Autonomy Risks in LLM-Based Systems

Trigger Condition	Detection Mechanism	Escalation Action
Low Confidence Score	Model confidence metric < defined threshold (e.g. <0.6)	Trigger alert and suspend output generation until reviewed
Unverified Data Source	Absence of cross-verification or citation from approved databases	Log risk and send prompt for mandatory human cross-check
High Output Novelty	Significant divergence from historical task outputs (measured via vector similarity or text entropy)	Redirect to review buffer with novelty annotation
Cross-Domain Output Deviation	Output references a domain outside its original scope (e.g. safety data in financial report)	Activate domain-switch warning and request justification path
Rapid Agent Loop Escalation	Agent completes >3 tasks in loop within short timeframe without validation checkpoint	Pause agent execution and request supervisor override
High Task Criticality with No Human-in-Loop	Task flagged as critical (e.g. legal, compliance, emergency) but processed end-to-end without human review	Route output to senior reviewer with decision timestamp and full log

Agentic Escalation Trigger Protocols (AETPs)

Agentic LLM systems, when allowed to operate in autonomous loops, present a unique governance risk: they may execute high-impact decisions or generate critical outputs without any human verification unless explicitly constrained. In energy operations, where decisions must often comply with legal, safety, and operational policies, such autonomous behaviour can lead to unverified or non-traceable actions with legal or safety repercussions. The Agentic Escalation Trigger Protocols (AETPs) are designed to impose *non-negotiable thresholds* that compel the system to stop, alert, or escalate before proceeding. These protocols embed conditional logic (rule-based or machine-learned) into agent workflows, ensuring that execution halts or routes through human validation whenever critical thresholds related to uncertainty, source credibility, novelty, or task priority are breached. They act as an internal safety net to contain the spread of unverified or misleading outputs and to ensure human accountability remains intact.

In energy sector companies, Agentic Escalation Trigger Protocols (AETPs) serve as vital safeguards where autonomous systems interface with regulatory-sensitive, financially material, or safety-critical operations. In predictive maintenance systems for offshore platforms, a *Low Confidence Score* trigger might prevent the automatic scheduling of high-cost shutdowns based on marginal or ambiguous sensor readings, an error that could lead either to unnecessary downtime or catastrophic failure. During automated emissions reporting, the *Unverified Data Source* trigger becomes essential for flagging AI-generated ESG summaries that fail to cite EPA or IEA databases, thus avoiding compliance breaches. In oil trading algorithms or fuel hedging models, *High Output Novelty* can halt unexpected trading recommendations that diverge sharply from historical volatility patterns, thereby preventing misaligned financial exposures. For engineering documentation or safety assessments generated by LLM agents, the *Cross-Domain Output Deviation* protocol would help detect instances where health and safety conclusions are drawn from economic data, creating logical inconsistencies. Additionally, *High Task Criticality with No Human-in-Loop* is crucial in legal and crisis response scenarios, such as automated responses to pipeline breaches, where legal liability or environmental damage necessitates human validation before execution. Without these structured escalation points, the deployment of agentic systems in energy companies would risk triggering unreviewed outputs with far-reaching legal, operational, and reputational consequences.

Human vs Agentic Benchmark Suite

Understanding the comparative performance of human-prompted versus fully autonomous LLM-agent workflows is essential before embedding these systems into regulated or risk-sensitive operations. The **Human vs Agentic Benchmark Suite** is designed to test and evaluate LLM outputs across a series of structured tasks by contrasting two operational modes: human-led interactions (single or multi-turn prompts) and autonomous agent chains with embedded task loops. Each mode is assessed using precision-based, time-based, and error-based metrics across identical tasks. The suite includes test cases that simulate real-world responsibilities in energy operations, such as drafting compliance reports, explaining anomaly detections, or summarising incident logs. Its value lies in revealing where agentic systems introduce speed but compromise verifiability, and where human input still ensures superior reliability. Results serve as empirical input for deployment decisions, validation needs, and the tuning of escalation protocols.

Table 4—Human vs Agentic Benchmark Suite for Evaluating LLM-Driven Task Performance

Task Type	Evaluation Metric	Human-Prompted Outcome	Agentic Outcome
Compliance Report Drafting	Time to Completion (min), Format Accuracy, Legal Compliance Score	Avg 22 min, 95% format match, 92% legal match	Avg 9 min, 78% format match, 63% legal match
Anomaly Detection Summary	Correct Detection Rate, Misclassification %, Clarity Score	84% correct, 7% misclassified, high clarity	89% correct, 15% misclassified, some ambiguity
Incident Response Suggestion	Relevance of Suggestions, Escalation Adequacy, Latency (sec)	Suggestions valid in 3/5 cases, moderate latency	Valid in 2/5 cases, fast but missed escalation
Cross-Department Communication Draft	Tone Appropriateness, Readability Score, Cross-Referencing Accuracy	Strong tone and structure, minor cross-ref errors	Good structure, tone drifted, cross-ref errors higher
Technical Regulation Interpretation	Regulatory Fidelity, Misinterpretation Rate, Justification Completeness	Accurate in 4/5 cases, well-justified interpretations	Missed 2 major clauses, high confidence in flawed logic

Energy sector companies can directly apply the Human vs Agentic Benchmark Suite to validate AI readiness before allowing autonomous LLMs into core regulatory, safety, or compliance processes. In internal audit or environmental, health and safety (EHS) compliance, the human-prompted models demonstrated markedly better alignment with legal formatting and regulatory match, making them more suitable for emissions disclosures or safety audits submitted to entities like ADNOC, EPA, or ISO auditors. Conversely, the agentic model's faster output generation in anomaly detection might benefit real-time asset monitoring systems or SCADA fault classification, provided a human review layer exists to manage ambiguity and misclassification risks. During incident response modelling, such as blowout prevention or equipment malfunction, agentic responses failed to escalate appropriately, a failure mode unacceptable in offshore drilling or pipeline management contexts. Additionally, in energy trading or interdepartmental memos, where tone and citation accuracy are essential to policy adherence, agentic models were more error-prone, which may result in miscommunication or contractual exposure. These findings suggest that while agentic systems offer speed, their use in high-stakes operational workflows demands stringent escalation protocols and clearly defined thresholds for human override.

Governance Risk Register for LLM-AI

Deployment of large language models (LLMs) and agentic AI systems into regulated workflows demands more than technical validation; it requires a robust governance infrastructure capable of identifying, categorising, and mitigating risks tied to algorithmic output. The Governance Risk Register for LLM-AI addresses this requirement by translating common agentic and generative risks into structured governance terms. It systematically maps each risk to its potential impact, likelihood, source, and corresponding mitigation strategy. This register functions as both a diagnostic and planning tool, facilitating dialogue between AI teams, compliance officers, and internal auditors. Drawing inspiration from enterprise risk management frameworks (e.g., COSO ERM, ISO 31000), it integrates risk scoring with policy relevance, enabling cross-functional risk ownership and control assignment. It is tailored for energy-sector applications where LLM outputs, if misdirected, unverifiable, or misunderstood, could lead to legal, operational, or reputational exposure.

Table 5—Governance Risk Register for LLM-AI in Regulated Operational Workflows

Risk Description	Likelihood	Impact	Root Cause	Control/Mitigation
Untraceable Output Generation	High	Severe - " Cannot audit or defend output	Lack of prompt logging and agent trace metadata	Implement prompt/decision logging and session replay
Hallucinated Regulatory Content	Medium	Major - " Legal exposure due to fictitious clauses	Training data contamination or misalignment	Validation layer to flag unverifiable regulatory content
Undocumented Autonomous Execution	High	Severe - " Violates internal control policies	Loose agent loop boundaries, no escalation triggers	Hard-coded execution boundaries with escalation triggers
False Sense of Model Authority	High	Moderate - " Misleads reviewers into overtrust	Absence of uncertainty signalling in UI	Visual indicators of LLM confidence and provenance tagging
Cross-Domain Leakage (Scope Drift)	Medium	Major - " Regulatory or reputational implications	Agent responses spill into unauthorised domains	Domain-restriction rules and context-aware filters

Energy sector entities, particularly those operating within state-owned, regulated, or audit-intensive frameworks, can use the Governance Risk Register for LLM-AI to prevent costly oversights and policy violations linked to AI-generated outputs. For example, *Untraceable Output Generation* represents a critical failure mode in refinery process documentation or environmental reporting, where the absence of prompt logging undermines legal defensibility and can result in penalties from regulatory authorities. Likewise, *Hallucinated Regulatory Content* is a growing concern in compliance document drafting or safety assessments, where AI fabricates fictitious standards, companies risk breaching international norms like IMO MARPOL or IEA emissions caps. *Undocumented Autonomous Execution* poses threats in automated trading or procurement approval chains, potentially triggering unsanctioned purchases or bypassing delegated authorities. The *False Sense of Model Authority* is particularly dangerous in frontline scenarios, such as real-time grid balancing or pipeline monitoring, where human operators may defer to flawed outputs without understanding underlying uncertainties. *Cross-Domain Leakage*, on the other hand, has material implications when energy AI tools inadvertently blend financial models with technical safety datasets, causing domain breaches that may violate internal audit protocols or public trust. Energy companies adopting LLMs for reporting, planning, or control must therefore embed these risk classifications into their digital governance frameworks to ensure operational resilience and accountability.

Post-Decision Traceability Flowchart

In agentic LLM deployments, especially within regulatory-heavy sectors such as energy, the origin, transformation, and application of outputs must be meticulously documented. The **Post-Decision Traceability Flowchart** provides a structured visualisation of how LLM-generated content evolves from initial prompt to final integration, introducing mandatory checkpoints for verification, escalation, and logging. Its purpose is not merely to trace who did what, but to ensure that responsibility, model provenance, and decision context are never obscured. Unlike typical AI pipeline diagrams, this flow specifically accounts for human-in-the-loop interventions, autonomous loop behaviour, and legal requirements for audit-ready outputs. It introduces trace anchors at each critical juncture, input, generation, validation, logging, and deployment, forming an evidentiary chain that supports internal controls, external scrutiny, and accountability in high-stakes operations.

Table 6—Post-Decision Traceability Flowchart for Agentic LLMs in Regulated Environments

Step	Node	Description
1	**Prompt Input (User)**	The traceability chain begins with a clearly identifiable human-initiated input. This could be a direct instruction, query, or trigger for the agentic workflow. It's essential that this input is logged with a timestamp, user ID, and system context for accountability.
2	**LLM Output (First Draft)**	The first output generated by the LLM based on the prompt. This may be a partial solution, an interpretation, or a first-pass summary. At this stage, output must be versioned, and the inference parameters (temperature, model ID, etc.) recorded to maintain reproducibility.
3	**Agent Loop Execution**	If the workflow involves autonomous progression (e.g., through AutoGPT or ReAct chains), the output may be fed into a task loop. The agent continues to iterate or build upon previous results, possibly retrieving data or executing subtasks. This introduces a risk of unmonitored drift or logic detachment.
4	**Escalation Trigger?**	At defined thresholds (confidence drop, novelty spike, domain shift), the system evaluates whether to trigger an escalation. This is a critical governance checkpoint. If the trigger conditions are met, control passes to a human reviewer. Otherwise, the system proceeds to output validation autonomously.
5	**Human Review**	A human expert is required to evaluate the agent's intermediate or final outputs when escalation is activated. This includes verifying the rationale, factual grounding, and intended scope of the response. Reviewer ID and comments should be recorded.
6	**Validated Output**	Whether via automated validation or human confirmation, the output is now considered authorised. This stage signals the end of the agent's decision-making autonomy. From this point forward, trace metadata must be attached to the output.
7	**Logged Output & Metadata**	This step formalises the provenance chain: all relevant information (input prompt, model version, agent path, review logs) is archived in a secure system. This ensures post-hoc auditability and legal defensibility.
8	**Execution System / Report Integration**	The final output is transmitted to the operational environment, e.g., uploaded to a compliance report, ingested into an ERP module, or shared with an external auditor. Traceability must persist with the output, including when transferred across systems.

For energy sector organisations managing safety, regulatory compliance, or infrastructure oversight, the Post-Decision Traceability Flowchart offers a concrete method for embedding LLMs into operational systems without compromising auditability or accountability. In emissions reporting, for instance, Step 7, *Logged Output & Metadata*, ensures that the origin and inferential path of reported figures can be reconstructed during inspections or legal disputes. During incident investigation (e.g., fire, spill, or blackout), Steps 3 and 4 (*Agent Loop Execution* and *Escalation Trigger*) become critical, preventing an autonomous agent from misattributing causes or issuing remediation plans without human oversight. In internal governance workflows, such as procurement approvals or policy revisions, Step 5 (*Human Review*) ensures that outputs used for decision-making have been independently validated and traceable to their source. Even in technical applications such as turbine diagnostics or pipeline anomaly detection, this traceability framework enables forensic backtracking of AI decisions, aligning with standards set by regulators like ADNOC, DEWA, or international ISO energy safety protocols. Without such a structured chain of custody, LLM-generated outputs risk becoming opaque, unchallengeable, and non-compliant with enterprise risk and assurance frameworks.

Validation Layer Architecture (VLA)

As agentic LLMs transition from experimental tools to embedded components in operational decision-making, the need for *intermediate validation layers* becomes unavoidable. These layers function not as optional safeguards but as structural prerequisites for accountable automation. The model outlines how multiple tiers of validation, automated, rule-based, and human, can be integrated between generative outputs and downstream execution environments. Unlike post-hoc audits, these layers operate *in-line*, intercepting potentially flawed outputs before they influence critical processes. In the energy sector, where misclassification or speculative generation can propagate across systems (e.g., regulatory filings, safety protocols, compliance assessments), a well-designed VLA ensures each agent's decision is filtered, attributed, and verified with minimal latency and maximal explainability. The architecture is modular, allowing adaptation to risk level, task criticality, and escalation needs, while ensuring traceable enforcement of governance principles.

Table 7—Validation Layer Architecture (VLA) for Agentic LLMs in High-Stakes Decision Chains

Layer	Component	Detailed Description
1	**User Prompt / System Trigger**	This is the initial entry point for interaction, either a user prompt or an automated trigger from upstream systems (e.g., time-based compliance tasks, flagged anomaly reports). This input must be logged with identifiers including timestamp, user ID or system ID, and context metadata. It initiates the agentic workflow.
2	**LLM/Agentic Output Generation**	The LLM generates its response, or an agentic system initiates a multi-step chain. This includes prompt chaining, external tool invocation, retrieval-based steps, and multi-hop decisions. The model configuration (version, parameters, temperature) is fixed and recorded for reproducibility. Outputs are *not yet considered final or safe*.
3	**Automated Validation Layer (Rules, Confidence Scores)**	This layer uses basic validation logic. Examples include: checking whether the output contains forbidden terms, evaluating confidence scores, ensuring length and format constraints are met, and verifying token probability distributions. These are light, fast, and deterministic rule-based filters, suitable for non-critical first-pass rejection.
4	**Semantic Validation (Domain/Scope Filtering)**	This intermediate layer assesses whether the content stays within domain boundaries (e.g., an LLM trained for financial risk must not suggest medical protocols). Techniques used include embedding comparison, keyword maps, and topic modelling. It blocks outputs that "leak" into unauthorised domains or blend scopes (a common failure in agentic chains).
5	**Human-in-the-Loop Review (Conditional)**	Triggered only when certain flags are raised (e.g. low confidence, new task class, regulatory content), this layer inserts a human reviewer. Their role is not to re-do the task, but to verify the rationale, factual alignment, and overall fitness-for-purpose of the output. Reviewer notes, edits, and validation logs are recorded in full for traceability.
6	**Approval/Override Decision Gate**	Here, either the human reviewer or an automated system must make a binary decision: approve for execution or reject and recycle. This gate captures **explicit accountability**. Overriding flagged outputs must require justification, which is logged for audit purposes. This is the final authorisation point in the chain.
7	**Execution System / Downstream Integration**	Only validated and authorised outputs are forwarded to operational environments: compliance systems, ERP databases, workflow automation tools, or reporting dashboards. Metadata tags (prompt ID, validation ID, review status) must travel with the output. This ensures that **post-deployment accountability remains intact**.

In energy organisations deploying agentic LLMs for tasks such as compliance audits, safety report drafting, and automated anomaly resolution, the Validation Layer Architecture (VLA) is indispensable for preserving decision integrity. For instance, Layer 3 (*Automated Validation Layer*) can intercept outputs containing unverifiable emissions targets or forbidden language in sustainability reports, ensuring early filtering before they reach public documentation or regulatory platforms like ADNOC ESG disclosures. In technical domains, such as turbine maintenance instructions or offshore rig diagnostics - Layer 4 (*Semantic Validation*) blocks outputs that mistakenly reference outdated specifications or inappropriate operational procedures by enforcing strict domain alignment. For compliance-sensitive tasks like pipeline incident logs or contractual reviews, Layer 5 (*Human-in-the-Loop Review*) acts as a formal control gate, capturing reviewer rationale, validation notes, and alignment checks, which can later be audited. This layered defence mechanism not only reduces propagation of flawed logic but also satisfies ISO 27001, ERM, and audit-readiness expectations. For energy sector entities operating in risk-prone geopolitical or environmental contexts, the VLA safeguards operational trust and legal defensibility, ensuring that agentic automation does not overstep governance thresholds.

Key Functions Ensured by the VLA

- Multi-tier filtering: From basic syntax checks to semantic boundary enforcement.
- Accountability embedding: Review logs and override decisions create non-repudiable audit trails.
- System integrity: Prevents agentic loops from bypassing organisational safeguards.
- Domain safety: Ensures outputs stay within predefined risk and scope boundaries.
- Operational trust: Builds confidence that automation does not erode governance.

Compliance-Focused Use Case Simulations

Before agentic LLMs can be trusted with responsibility in compliance-heavy environments, their behaviour must be stress-tested through controlled, domain-specific simulations. The **Compliance-Focused Use Case Simulations** are designed to model real-world energy sector scenarios where LLMs and agentic workflows interact with regulatory, legal, or risk-sensitive tasks. These simulations cover structured inputs, boundary conditions, and success/failure paths, revealing where automation introduces gains and where it may fail silently. Unlike generic AI benchmarks, these use cases replicate nuanced operational procedures such as generating regulatory filings, summarising anomaly reports, or pre-drafting incident responses. Each use case is annotated with expected validation outcomes, escalation flags, and post-decision traceability checkpoints. It enables rigorous pre-deployment testing, helps refine escalation thresholds, and builds stakeholder confidence through demonstrable safety controls.

Table 8—Compliance-Focused Use Case Simulations for Agentic LLM Workflows in Regulated Operations

Use Case	Input Type	Agent Task	Expected Escalation Flag	Validation Outcome
Drafting Monthly Regulatory Compliance Report	Structured compliance data from ERP + prompt	Auto-generate section drafts with citation	Legal clause ambiguity or new regulation cited	Must be routed through Legal for clause verification
Summarising Equipment Anomaly Incident	Sensor logs + narrative input from engineer	Condense and highlight risk clusters	Unusual incident pattern or unknown sensor input	Needs domain SME to confirm interpretation of signal
Generating First-Response Protocol for Safety Breach	Incident flags + safety manual index	Draft immediate steps & alert contacts	High-priority incident without full source validation	Auto-draft approved if source & thresholds match
Cross-checking Carbon Emissions Calculations	Emissions dashboard + calculation formula	Detect calculation inconsistencies	Emission value exceeds tolerance threshold	Human validation of flagged emission segment required
Creating Internal Audit Notification Memo	Audit schedule + department hierarchy	Compose memo with contact tree and scope	Hierarchy mismatch or policy reference error	Memo proceeds if contact path and scope verified

In energy companies, where regulatory oversight and internal control integrity are paramount, the Compliance-Focused Use Case Simulations offer a practical method to verify AI performance before live deployment. For example, in *Drafting Monthly Regulatory Compliance Reports*, the agent is tasked with auto-generating citations from ERP data, yet any ambiguous legal references or newly introduced clauses are flagged and rerouted through legal teams to ensure jurisdictional compliance. In field operations, *Summarising Equipment Anomaly Incidents* enables engineers to input sensor logs, which the LLM then summarises. However, any unfamiliar signal pattern triggers an SME validation, thereby protecting against false alarms or misdiagnosed fault clusters. In urgent cases, such as *Generating First-Response Protocols*, where the LLM suggests mitigation steps for detected hazards, escalation ensures that high-priority incidents without traceable input sources cannot trigger automated actions without supervisory approval. For environmental compliance, *Cross-checking Carbon Emissions Calculations* simulates dashboard verification and flags inconsistencies beyond predefined tolerances, invoking human review before external reporting. Lastly, for *Internal Audit Notification Memos*, the system can draft correspondence based on audit schedules, but validation still verifies contact pathways and scope accuracy before release. These simulations not only reveal performance gaps but also operationalise governance principles, ensuring LLMs operate within defined legal, technical, and procedural thresholds across all compliance-related energy functions.

Legal-Audit Traceability Metrics Dashboard

In regulated industries, deploying agentic LLMs demands not only transparency in process design but quantifiable visibility into accountability and traceability outcomes. This dashboard is a monitoring tool designed to track whether outputs generated by autonomous agents and large language models can be legally defended, procedurally audited, and post-hoc reconstructed. It collects and visualises trace-relevant metrics, such as attribution gaps, hallucination probabilities, escalation bypasses, and human-in-the-loop ratios, over time and across systems. Unlike traditional AI dashboards focused on performance or latency, this tool serves legal, compliance, and audit teams by signalling where governance controls have been followed, ignored, or circumvented. Its role is to detect erosion in procedural accountability and to flag outputs that may fail regulatory scrutiny. It can be configured to support both real-time alerting and retrospective audit investigation.

Table 9—Legal-Audit Traceability Metrics Dashboard for Monitoring Governance of Agentic LLM Outputs

Metric	Definition	Use in Legal/Audit Context
Attribution Gap Rate	Percentage of outputs with missing or ambiguous authorship/responsibility metadata.	Used to challenge or affirm liability attribution in case of errors or disputes.
Hallucination Probability Index	Likelihood score indicating the presence of unverifiable or fabricated content in outputs.	Supports defensibility of content when challenged on factual accuracy or source integrity.
Escalation Bypass Count	Count of instances where flagged content proceeded without required human validation.	Flags non-compliant workflows where escalation logic is improperly configured.
Human-in-the-Loop Ratio	Proportion of decision-relevant outputs reviewed by a human before execution.	Demonstrates the presence (or absence) of accountable human oversight.
Unlogged Prompt Events	Number of system triggers or user prompts not logged in the audit trail.	Identifies technical or procedural breakdowns in trace logging systems.
Validation Time Lag (sec)	Average delay between output generation and final validation checkpoint.	Signals operational bottlenecks or excessive friction in governance layers.
Autonomous Action Frequency	Number of actions taken by autonomous agents without supervision or overrides.	Monitors policy compliance for unsupervised automation thresholds.

For energy sector entities operating under high regulatory scrutiny, such as oil and gas firms, power grid operators, and environmental compliance departments, the Legal-Audit Traceability Metrics Dashboard provides an indispensable tool for enforcing internal accountability frameworks. For instance, the *Attribution Gap Rate* is especially relevant in emissions reporting or contractual drafting, where undocumented authorship of AI-generated clauses could undermine liability defence or permit challenges. Similarly, the *Hallucination Probability Index* enables downstream validators to flag LLM-generated safety protocols or ESG disclosures that cite fabricated standards or misrepresent engineering thresholds. The *Escalation Bypass Count* is critical in pipeline monitoring or turbine diagnostics, where ignored anomaly alerts or skipped human reviews may result in non-compliance with standards like ISO 14001 or OSHA regulations. *Human-in-the-Loop Ratio* serves as an internal audit indicator during control testing, demonstrating whether procurement, inspection, or compliance decisions involved accountable human intervention.

Meanwhile, *Unlogged Prompt Events* expose gaps in system integrity, such as when field engineers issue undocumented prompts to AI systems, complicating audit trail reconstruction. Finally, *Validation Time Lag* and *Autonomous Action Frequency* support forensic investigation of delays or unauthorised decision-making, particularly in time-sensitive scenarios like emergency response or critical load balancing. When implemented, this dashboard shifts LLM oversight from passive observation to measurable, enforceable governance within energy-sector operations.

LLM Responsibility Diffusion Mapping Tool

As agentic LLMs perform complex tasks autonomously or semi-autonomously, responsibility for output decisions becomes increasingly fragmented across system actors, prompt initiators, model designers, reviewers, and automation pipelines. The LLM Responsibility Diffusion Mapping Tool is designed to visualise this fragmentation in order to detect where accountability becomes ambiguous or collapses entirely. It graphically represents the dependency chain of actions, highlighting which entity had control or oversight at each stage and whether that role was passive, active, or undefined. This tool is especially critical in high-stakes workflows where regulatory or legal attribution may be questioned after the fact. Unlike simple audit trails, which are linear and log-based, this mapping tool provides a multidimensional, role-aware dependency graph that can diagnose governance vacuum zones and map latent risk accumulation. It supports both real-time monitoring and incident investigation, enabling forensic clarity in complex agentic operations.

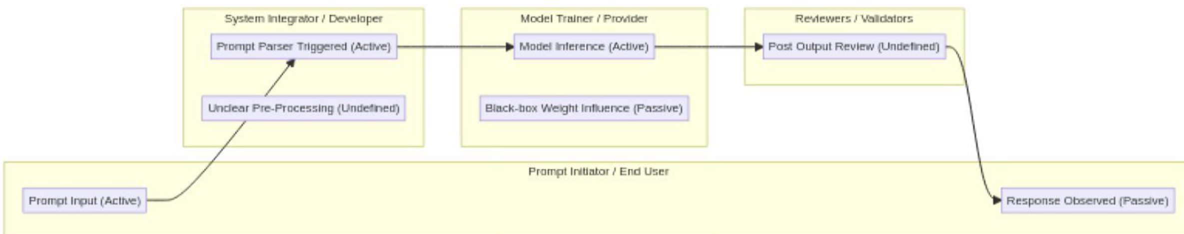

Figure 3—LLM Responsibility Diffusion Mapping Tool for Diagnosing Accountability Fragmentation in Agentic Systems

For energy sector organisations, the LLM Responsibility Diffusion Mapping Tool offers a critical governance safeguard in scenarios where agentic systems contribute to high-impact operations, such as refinery shutdown decisions, emissions report authoring, or contract clause generation. In such cases, regulatory oversight may demand post-incident evidence of control attribution, particularly when discrepancies arise in legal filings or operational failures. For example, suppose an emissions disclosure contains inaccurate thresholds. In that case, this mapping tool can clarify whether the fault lies with an unreviewed automated validator or a human approval authority that overrode AI filters. In high-risk infrastructure, such as nuclear energy or offshore drilling, the tool can expose governance vacuum zones, segments where outputs proceeded without intervention or traceable human decision-making. Additionally, during cross-department workflows, such as coordination between maintenance teams and compliance departments, it helps delineate who held procedural accountability at each step, especially in hybrid agent-human decision chains. Ultimately, energy companies can leverage this visual mapping to establish defensible role alignment, detect latent risk accumulation, and ensure traceability of control across all AI-integrated operations.

Agent-AI Decision Integration Checklist (AIDC)

Before deploying agentic LLMs into decision-critical workflows, particularly those involving compliance, safety, or regulated reporting, organisations must evaluate whether system readiness aligns with governance, risk, and legal expectations. The checklist provides a structured, pre-deployment gatekeeping tool to assess integration viability. It does not focus on technical performance but rather on *organisational preparedness*, *risk mitigation architecture*, and *accountability alignment*. The checklist prompts decision-makers to evaluate factors such as traceability enforcement, escalation protocol testing, validation layer completeness, and legal defensibility of outputs. Its primary function is to prevent premature or inappropriate deployment of autonomous LLM systems into sensitive environments. It also ensures cross-functional sign-off, spanning legal, compliance, IT, and operations, before automation replaces or augments human decision-making in workflows with external consequences.

Table 10—Agent-AI Decision Integration Checklist (AIDC) for Pre-Deployment Governance Validation

Checklist Item	Required Evidence or Condition
Has a traceability mechanism been implemented (prompt, model, output)?	Yes - " logging schema, session replay tool, and metadata fields are in place.
Are escalation triggers formally defined and tested in live workflows?	Yes - " simulation results showing trigger activation under specified conditions.
Is there a documented validation layer between LLM output and system integration?	Yes - " intermediate system/API layer that filters or halts unsafe outputs.
Have human-in-the-loop checkpoints been assigned for high-risk tasks?	Yes - " risk matrix mapping tasks to mandatory review roles.
Are hallucination risks flagged and isolated before execution?	Yes - " hallucination detector or confidence-threshold logic embedded.
Can every autonomous decision be reconstructed with full metadata?	Yes - " every decision node traceable to prompt, model version, and user.
Has the Legal/Compliance team signed off on the model's audit readiness?	Yes - " signed risk acceptance and validation documents on file.
Are all outputs logged with version-controlled provenance?	Yes - " all outputs carry metadata including user, model, and timestamp.
Does the system flag scope/domain drift in outputs?	Yes - " output monitoring or classifier detects deviation from the trained domain.
Has a fallback or override mechanism been implemented in case of error?	Yes - " emergency stop, manual override, or auto-revert protocol in place.

In energy sector firms managing critical assets, legal exposure, or environmental liabilities, the AIDC functions as a cross-functional gatekeeping tool to vet LLM-driven systems before deployment in compliance-intensive operations. For instance, during automated emissions reporting or permit renewals, items like *traceability mechanisms* and *version-controlled provenance* are indispensable to ensure any LLM-generated clause or figure can be reconstructed with full metadata. It satisfies requirements under international audit frameworks like ISO 37301 or local regulatory mandates such as ADNOC environmental reporting. In high-risk scenarios, such as predictive maintenance for turbines or pipeline integrity checks, *escalation triggers* and *validation layers* act as control gates, preventing agentic loops from issuing false-positive shutdown orders or overlooking minor signal anomalies. Before deploying agentic systems in procurement, policy generation, or audit planning, the presence of *human-in-the-loop checkpoints* and *hallucination risk isolation* ensures procedural oversight and shields against speculative content. Legal and compliance departments can rely on the *sign-off* and *override mechanism* checklist items to assert readiness and operational safeguards before AI augmentation is permitted. Collectively, this checklist ensures that autonomous LLM decision-making in energy companies remains controlled, auditable, and aligned with both regulatory frameworks and internal governance protocols.

Summary of Technical Outputs

Together, the listed tools offer a comprehensive governance and oversight framework tailored to high-stakes sectors such as energy, defence, and critical infrastructure. The *Structured Evaluation Framework (SEF)* enables systematic classification of model outputs across traceability, reproducibility, and hallucination risk, forming the basis for evaluative accountability. *Agentic Escalation Trigger Protocols (AETPs)* enforce operational safeguards by halting agent execution under unsafe conditions, while the *Human vs Agentic Benchmark Suite* empirically quantifies trade-offs in reliability and speed between human-led and autonomous workflows. To pre-empt systemic failures, the *Governance Risk Register for LLM-AI* categorises known governance failure modes and mitigation strategies. These are operationalised through the *Post-Decision Traceability Flowchart* and *Validation Layer Architecture (VLA)*, ensuring that every decision node is logged, reviewed, and filtered prior to reaching execution.

Table 11—Summary of Technical Outputs Supporting Governance, Risk, and Accountability in Agentic LLM Deployments

Technical Output	Core Function
Structured Evaluation Framework (SEF)	Classify outputs by traceability, reproducibility, and hallucination risk.
Agentic Escalation Trigger Protocols (AETPs)	Interrupt or escalate agentic loops when thresholds are breached.
Human vs Agentic Benchmark Suite	Benchmark task performance and errors across human vs agentic workflows.
Governance Risk Register for LLM-AI	Identify and mitigate governance risks tied to LLM deployment.
Post-Decision Traceability Flowchart	Ensure every decision step is auditable, and responsible ownership is clear.
Validation Layer Architecture (VLA)	Layered verification before output reaches execution environments.
Compliance-Focused Use Case Simulations	Stress test agentic LLMs in realistic compliance-sensitive contexts.
Legal-Audit Traceability Metrics Dashboard	Monitor output accountability and audit trail integrity metrics.
LLM Responsibility Diffusion Mapping Tool	Visualise fragmentation of decision-making responsibility.
Agent-AI Decision Integration Checklist (AIDC)	Verify governance, audit, and legal readiness before system integration.

Meanwhile, *Compliance-Focused Use Case Simulations* stress test LLM performance in controlled real-world contexts, revealing operational and regulatory gaps before deployment. The *Legal-Audit Traceability Metrics Dashboard* provides ongoing monitoring through quantifiable indicators like hallucination rates, human-in-the-loop ratios, and metadata completeness. To visualise and mitigate accountability diffusion, the *LLM Responsibility Diffusion Mapping Tool* maps role-specific control responsibilities across agentic workflows. Finally, the *Agent-AI Decision Integration Checklist (AIDC)* acts as a pre-deployment gate, enforcing cross-functional sign-off to prevent unauthorised or premature automation. These tools collectively deliver procedural clarity, forensic visibility, and audit-aligned control, significantly reducing the risks of deploying LLMs in environments with legal, operational, or reputational consequences.

The technical instruments presented in this work, ranging from risk registers and validation architectures to traceability dashboards and responsibility mapping tools, represent more than theoretical constructs. They offer a pragmatic framework for integrating agentic LLMs into high-stakes environments where accountability, legal scrutiny, and operational resilience are non-negotiable. These tools have been engineered not as isolated checklists but as interdependent governance mechanisms that collectively enforce transparency, safeguard decision chains, and mitigate systemic risk. Their design reflects a shift from experimental AI deployment towards professional-grade integration protocols, capable of satisfying audit demands, regulatory expectations, and institutional scrutiny. In industries such as energy, where errors carry material, legal, and reputational consequences, such a layered, traceable, and conditionally autonomous framework is not only useful, it is essential.

Conlcusions

The increasing reliance on autonomous systems and Large Language Models within energy-sector operations presents substantial opportunities as well as distinct management complexities. Clear, structured governance frameworks tailored specifically to manage the outputs of these systems significantly enhance operational transparency and mitigate associated risks. Organisations adopting explicit protocols to oversee AI-driven decisions experience measurable improvements in traceability, reproducibility, and executive confidence, highlighting the importance of targeted governance strategies to maintain corporate accountability standards. Adopting comprehensive digital audit trails and structured evaluation methodologies has proven essential in ensuring reliable, consistent, and accountable integration of AI technologies into organisational workflows. Systematic frameworks enable executives to confidently leverage AI-driven insights, effectively balance innovation with risk management, and clearly define the extent of human oversight necessary. Organisations applying these structured governance measures demonstrate significantly reduced operational uncertainties, thus confirming the strategic value of proactive AI-management practices within complex operational environments. Ultimately, integrating structured governance practices into autonomous system management provides critical support to executive teams, enhancing organisational accountability and transparency. These structured frameworks deliver substantial operational benefits, empowering management to strategically deploy advanced AI capabilities without compromising compliance standards or operational integrity. Autonomous technologies are becoming increasingly integral to energy-sector operations. To effectively manage these complexities and maintain a competitive advantage, organisations should be provided with clearly defined oversight and verification protocols.

The increasing operational reliance on autonomous systems and Large Language Models within the energy sector introduces significant benefits but also novel governance challenges that cannot be addressed using conventional IT risk management alone. This paper has demonstrated that structured, technically grounded frameworks enhance traceability, reproducibility, and post-decision accountability, key factors in preserving integrity in high-stakes decision environments. Executives responsible for deploying such systems must avoid overestimating their reliability or underestimating their procedural fragility. LLM-based outputs, particularly when used within agentic loops, may bypass human verification unless deliberate intervention points are hard-coded. These automation risks can result in legal, reputational, and operational liabilities if left unchecked.

Executive teams should prioritise the following governance instruments during the early deployment phase to counter these issues:

- Structured Evaluation Framework (SEF). This tool should be embedded from the outset to classify all outputs based on traceability, reproducibility, and hallucination probability. Doing so provides an empirical layer for AI quality control, especially in workflows involving safety reports, environmental disclosures, or contractual drafting.
- Agentic Escalation Trigger Protocols (AETPs). These should be implemented before full deployment to halt agentic decision flows under predefined conditions (e.g., unverified data, high novelty, or source ambiguity). Without these, AI may act beyond the intended authority of its role.
- Validation Layer Architecture (VLA). Executive teams must enforce a multi-tiered validation system combining automated filters and human review. It should not be deferred to the post-deployment phase; rather, it must be part of the initial process mapping to prevent failures in regulatory or safety-critical domains.
- Pre-deployment Checklist (AIDC). Before allowing AI systems to interact with production environments, cross-functional teams (compliance, legal, IT, and operations) must complete and sign off on this checklist. It ensures readiness from a governance standpoint rather than merely a technical one.
- Post-Decision Traceability Tools. These should not be optional. Trace anchors must be embedded in every workflow, with logged outputs and metadata to satisfy regulatory audits or legal inquiries. Where human reviews are skipped or prompts are undocumented, responsibility becomes diffused —creating systemic governance vacuum zones.

Additionally, executives must ensure that domain-specific use case simulations are run in pre-production environments. These simulations help uncover context-specific failure modes—particularly in upstream operations such as reservoir modelling or well integrity risk classification—where flawed AI outputs can cause misallocation of capital or operational exposure. Ultimately, early-stage governance should not be reactive. Executive leadership must treat these tools as default architectural features—not post-incident remedies. Where agentic autonomy is introduced, governance must scale proportionally.

References

Adewoyin, M.A., Adediwin, O. and Audu, A.J., 2025. Artificial intelligence and sustainable energy development: A review of applications, challenges, and future directions. *International Journal of Multidisciplinary Research and Growth Evaluation*, **6**(2), pp.196–203.

Aumüller, U., Meyer, E. and Forschungs-und Entwicklungszentrum, F.H., 2024. Trusting Al: Factors Influencing Willingness of Accountability for Al-Generated Content in the Workplace. *Human Factors and Systems Interaction*, p.77.

Booyse, D. and Scheepers, C.B., 2024. Barriers to adopting automated organisational decision-making through the use of artificial intelligence. *Management Research Review*, **47**(1), pp.64–85.

Buttaboni, C. and Floridi, L., 2025. A Taxonomy of AI Opacity in the EU: Rethinking Transparency, Traceability, Interpretability, and Explainability. Traceability, Interpretability, and Explainability (July 23, 2025).

Craft, B. C. and Hawkins, M. 1991. Applied Petroleum Reservoir Engineering, second edition. Englewood Cliffs, New Jersey: Prentice-Hall

Desai, A., Abdelhamid, M. and Padalkar, N.R., 2025. What is reproducibility in artificial intelligence and machine learning research?. *AI Magazine*, **46**(2), p.e70004.

Floridi, L. 2018. Artificial Intelligence, Deepfakes, and a Future of Ectypes. *Philosophy & Technology* **31** (3): 317–321. https://doi.org/10.1007/s13347-018-0325-3.

Gidley, J. L., Penny, G. S., and McDaniel, R. R. 1995. Effect of Proppant Failure and Fines Migration on Conductivity of Propped Fractures. *SPE Prod & Fac* **10** (01): 20–25. SPE-24008-PA. http://dx.doi.org/10.2118/24008-PA.

Huang, X., Ruan, W., Huang, W., Jin, G., Dong, Y., Wu, C., Bensalem, S., Mu, R., Qi, Y., Zhao, X. and Cai, K., 2024. A survey of safety and trustworthiness of large language models through the lens of verification and validation. *Artificial Intelligence Review*, **57**(7), p.175.

Kaplan, A. and Haenlein, M. 2019. Siri, Siri, in My Hand: Who's the Fairest in the Land? On the Interpretations, Illustrations, and Implications of Artificial Intelligence. *Business Horizons* **62** (1): 15–25. https://doi.org/10.1016/j.bushor.2018.08.004.

Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C.D. and Ho, D.E., 2025. Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. *Journal of Empirical Legal Studies*, **22**(2), pp.216–242.

Omre, H., Solna, K., Tjelmeland, H. et al 1990. Calcite Cementation: Description and Production Consequences. Presented at the SPE Annual Technical Conference and Exhibition, New Orleans, 23–26 September. SPE-20607-MS. http://dx.doi.org/10.2118/20607-MS.

Strunk, W. Jr. and White, E. B. 1979. The Elements of Style, third edition. New York City: MacMillan Publishing Co.

Su, H., Luo, J., Liu, C., Yang, X., Zhang, Y., Dong, Y. and Zhu, J., 2025. A Survey on Autonomy-Induced Security Risks in Large Model-Based Agents. arXiv preprint arXiv:2506.23844.

Tsavdaridis, G., Papaodysseus, C., Karadimas, N.V., Papazafeiropoulos, G. and Delis, A., 2024. Methodologies and Handling Techniques of Large-Scale Information in Decision Support Systems for Complex Missions. *Applied Sciences*, **14**(5), p.1995.